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Automation techniques

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Automation techniques

01	Automation patterns and options for pipelines
02	Cloud Scheduler and Workflows
03	Managed Service for Apache Airflow
04	Cloud Run functions
05	Eventarc



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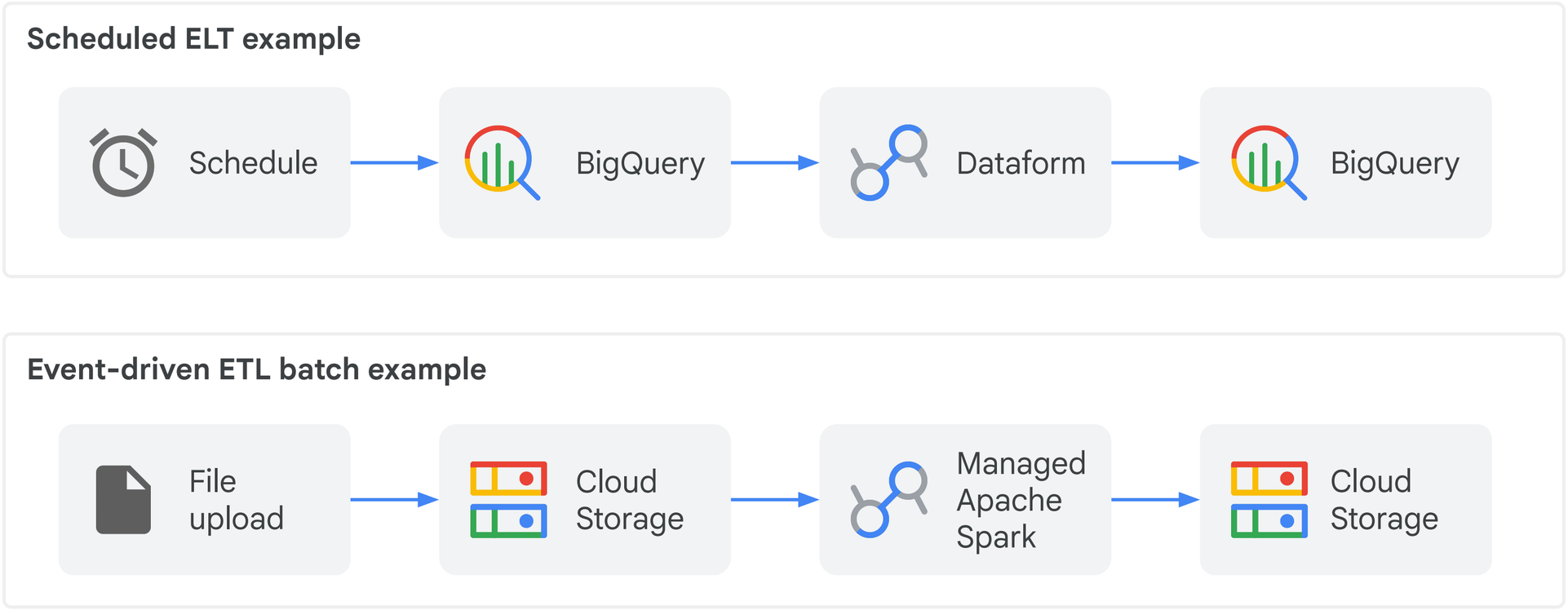
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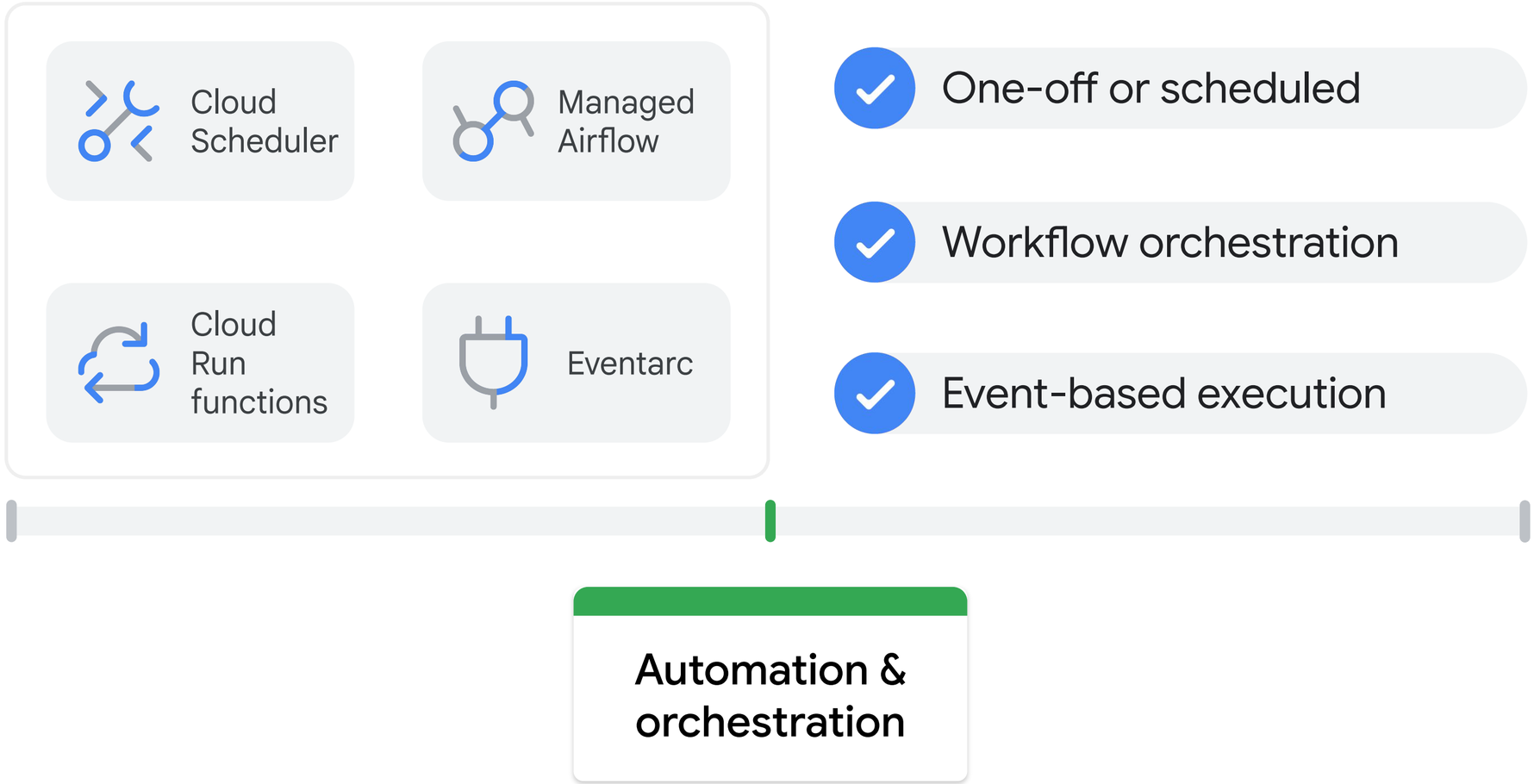
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ELT and ETL workloads can be automated to run on a recurring basis



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Google Cloud provides multiple services for automating and orchestrating your workloads



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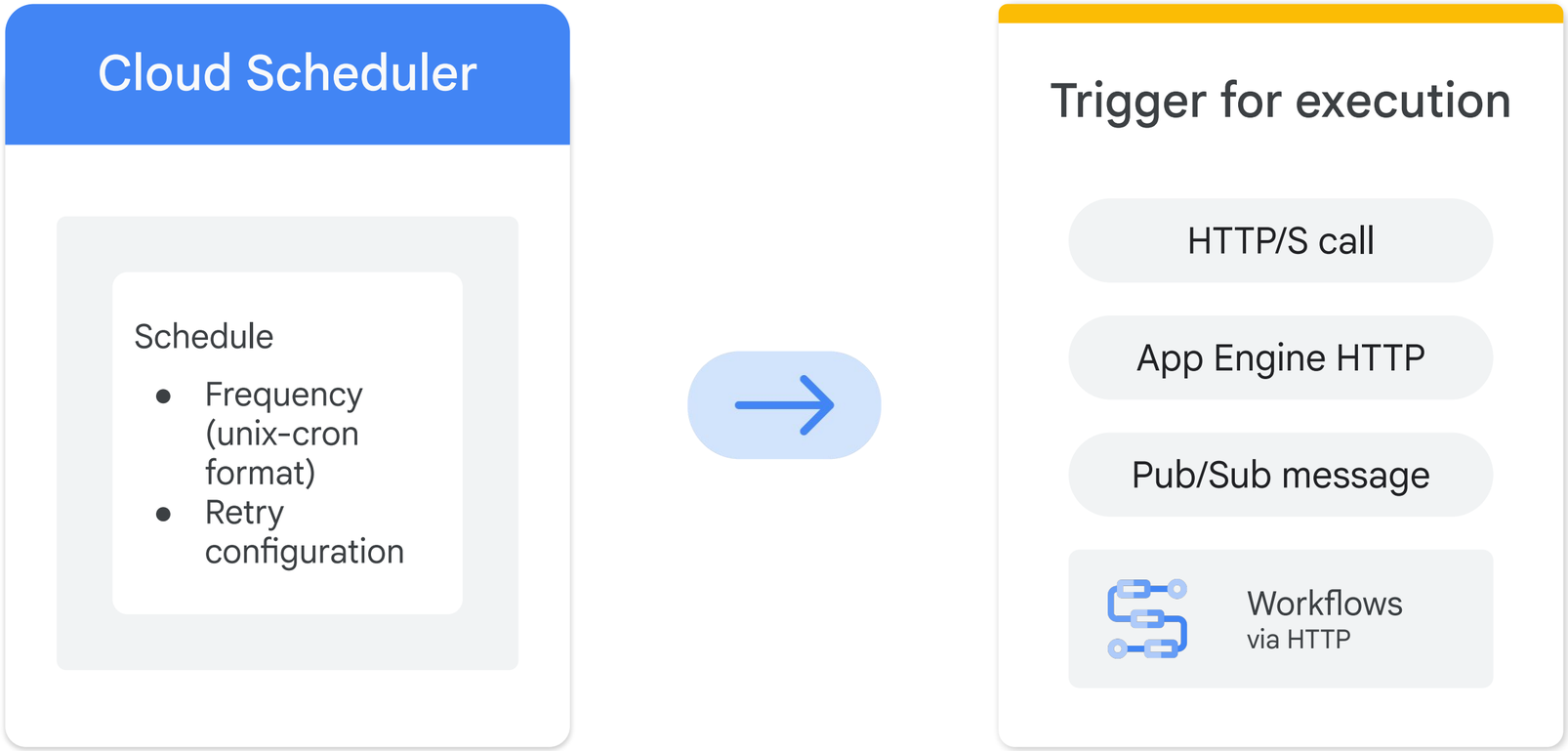
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Cloud Scheduler invokes your workloads at recurring intervals



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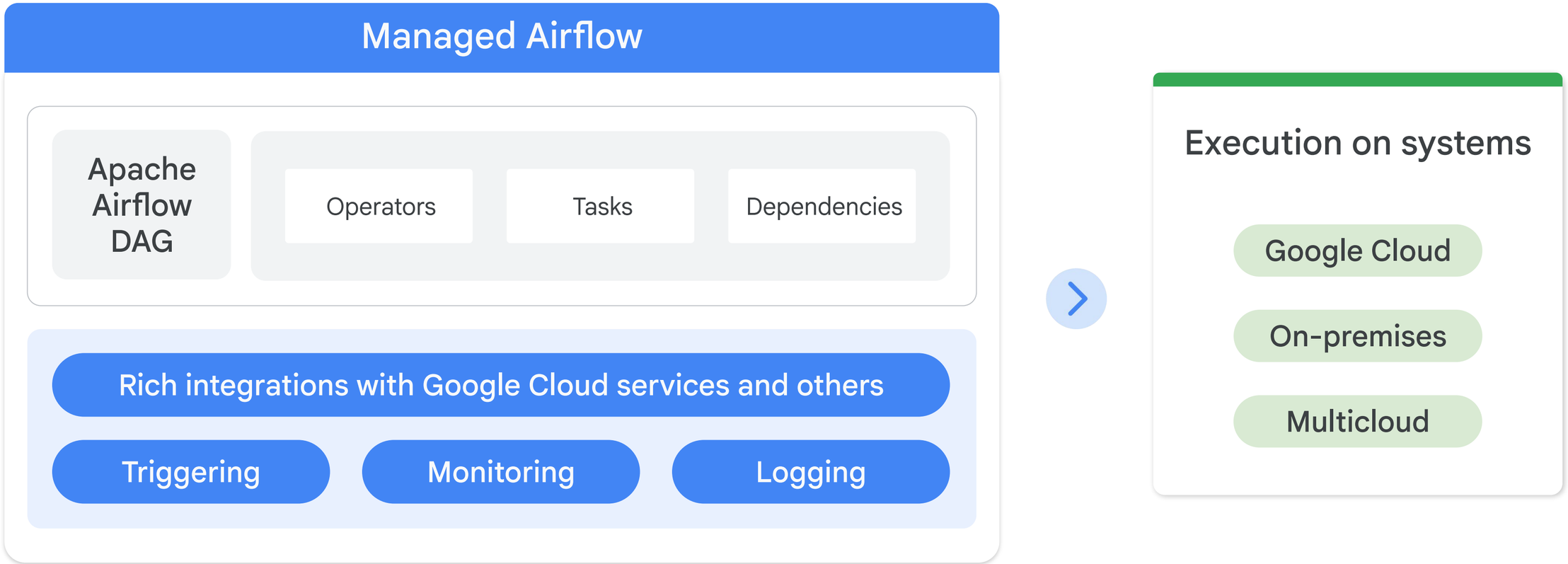
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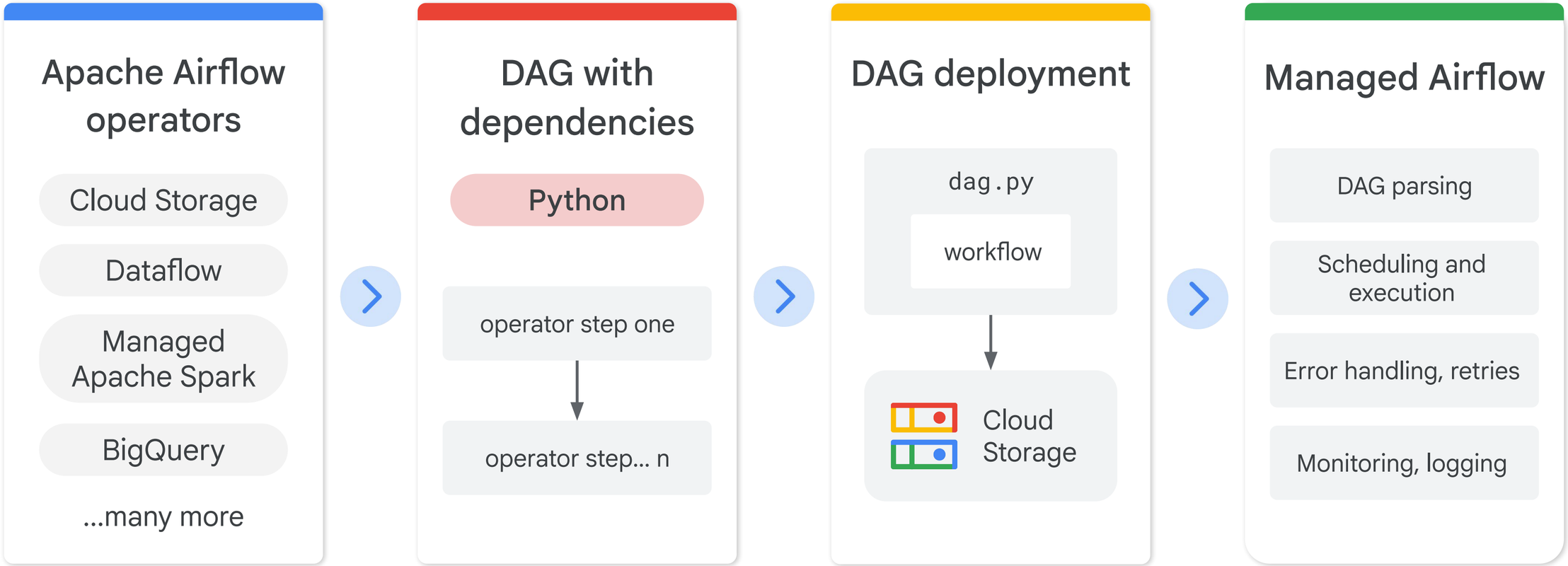
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Managed Airflow orchestrates your pipelines on different systems into workflows



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Develop your DAG in Python using Apache Airflow operators and submit it to Managed Airflow for execution



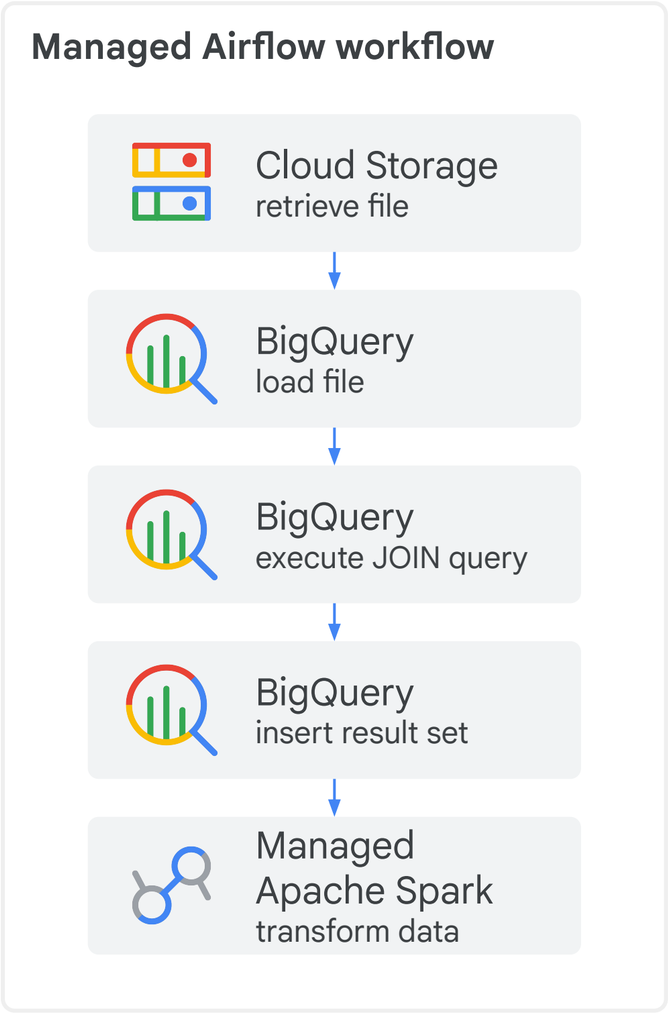
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Example: Run a data analytics DAG

```
with models.DAG(
    "data_analytics_dag",
    # Define schedule and default args
) as dag:
    create_batch = DataprocCreateBatchOperator(
        # Specify Managed Service for Apache Spark settings, e.g. which
        # Python file to execute
    )
    load_external_dataset = GCSToBigQueryOperator(
        # Specify Cloud Storage source file to load
        # and BigQuery table destination
    )
    with TaskGroup("join_bq_datasets") as bq_join_group:
        # Define the SQL query in BigQuery to join
        # the loaded table with another one
            bq_join_holidays_weather_data = BigQueryInsertJobOperator(
                # Execute query and insert result into BigQuery table
            )

        # define the dependencies of the workflow
        load_external_dataset >> bq_join_group >> create_batch
```





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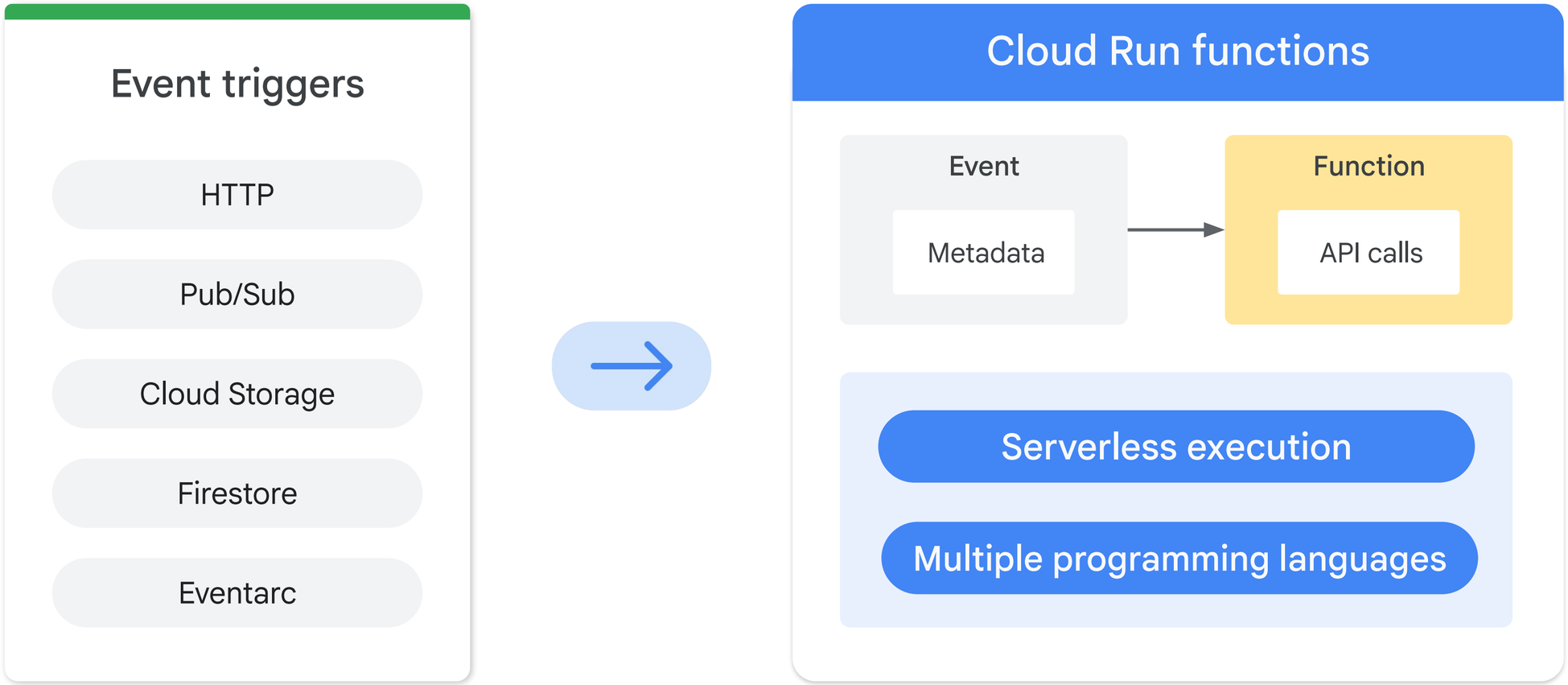
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Use Cloud Run functions to execute code based on Google Cloud events



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Example: Trigger a Managed Service for Apache Spark workflow template after a file upload to Cloud Storage

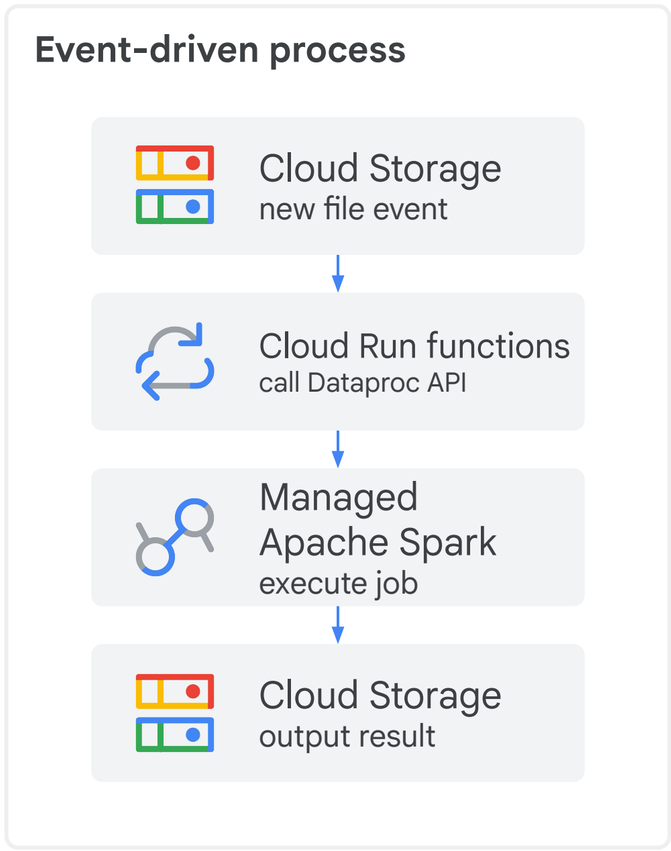
```
// pre-work: define project ID, workflow template, region
// set up Dataproc API client in specific region
const client = new dataproc.WorkflowTemplateServiceClient({
  apiEndpoint: `${region}-dataproc.googleapis.com`,
});

// retrieve bucket and name of new object on Cloud Storage
const file = data;
const inputBucketUri = `gs://${file.bucket}/${file.name}`;

// construct request to Dataproc API
const request = {
 name: client.projectRegionWorkflowTemplatePath(projectId, region, workflowTemplate),
 parameters: {"INPUT_BUCKET_URI": inputBucketUri}
};

// call API to launch the workflow
client.instantiateWorkflowTemplate(request)
 .then(responses => {console.log("Launched Dataproc Workflow:", responses[1]);
 })
 .catch(err => {console.error(err);
 });
```





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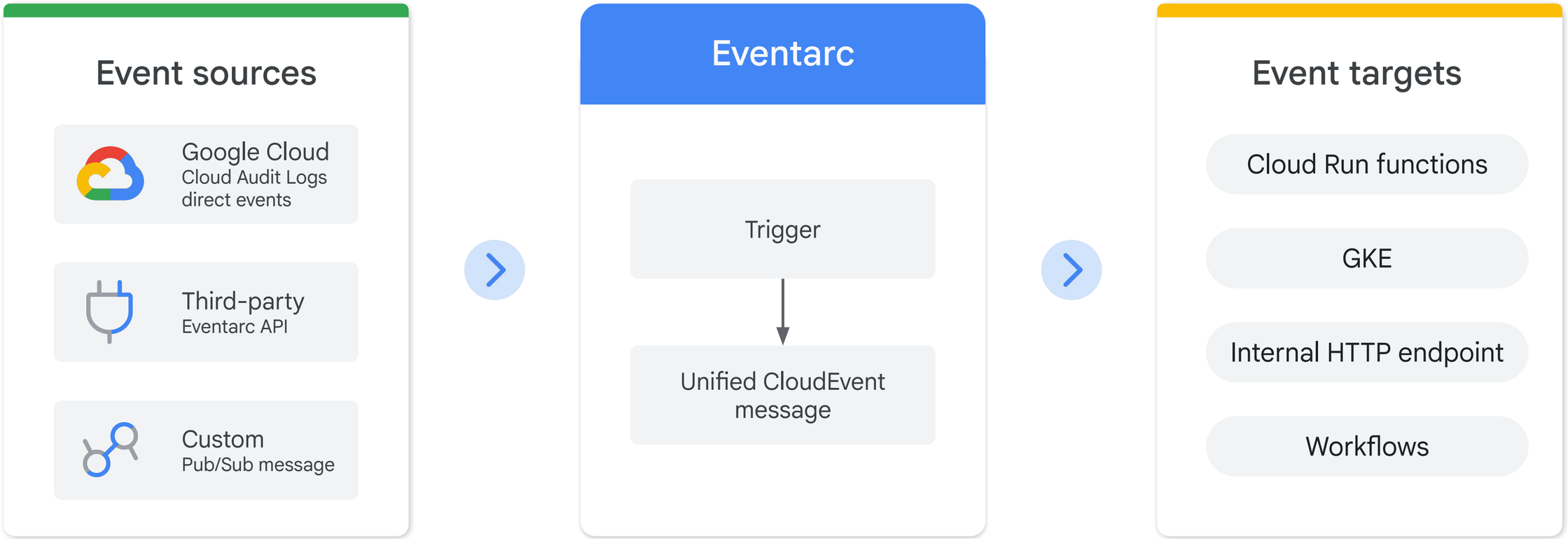
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Build a unified event-driven architecture for loosely coupled services with Eventarc



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Example: Respond to `INSERT` events in BigQuery

Cloud Audit Logs event as Eventarc trigger:

Infrequent event

`INSERT`



Table
BigQuery

```
{
  protoPayload: {
    @type: "type.googleapis.com/google.cloud.audit.AuditLog"
    metadata: {
      @type:
        "type.googleapis.com/google.cloud.audit.BigQueryAuditMetadata"
    tableDataChange: {
      insertedRowsCount: "2"
      jobName: "projects/<PROJECT>/jobs/bquxjob_4d3da71c_190de71a2f3"
      reason: "QUERY"
    }
  }
  methodName: "google.cloud.bigquery.v2.JobService.InsertJob"
  resourceName: "projects/<PROJECT>/datasets/<DATASET>/tables/<TABLE>"
  serviceName: "bigquery.googleapis.com"
}
```

Execution



Rebuild dashboard
Cloud Run



Retrain ML model
Cloud Run

...other custom action

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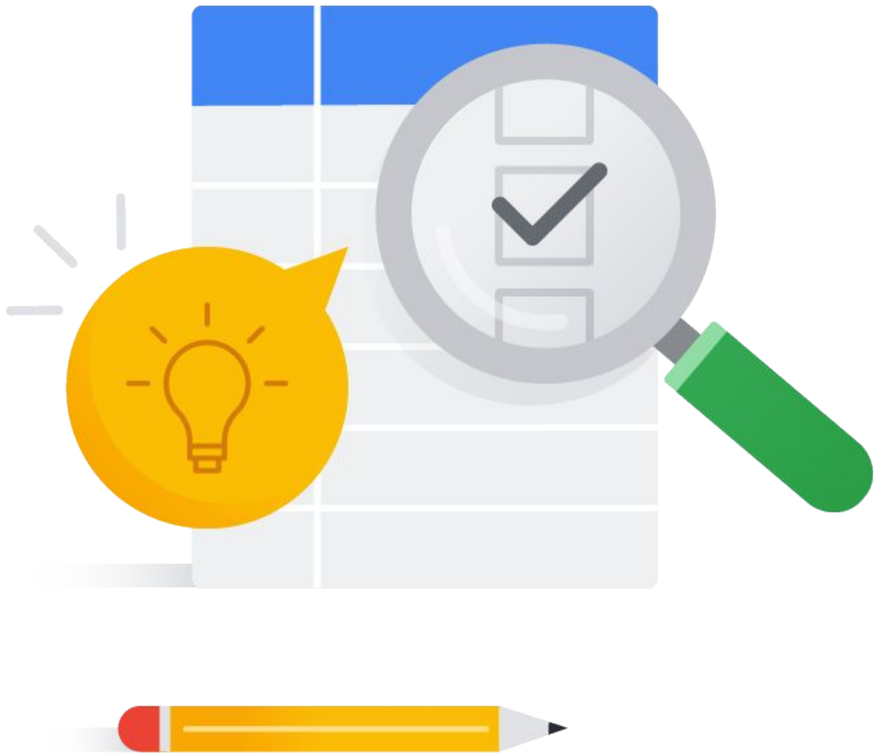
Compare automation options

	Cloud Scheduler	Managed Airflow	Cloud Run functions	Eventarc
Trigger type	schedule, manual	schedule, manual	event	event
Serverless	yes	no	yes	yes
Coding effort	low	medium	high	high
Programming languages	YAML (with Workflows)	Python	Python, Java, Go, Node.js, Ruby, PHP, .NET core	any

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Lab (optional): Use Cloud Run Functions to Load BigQuery

30 min

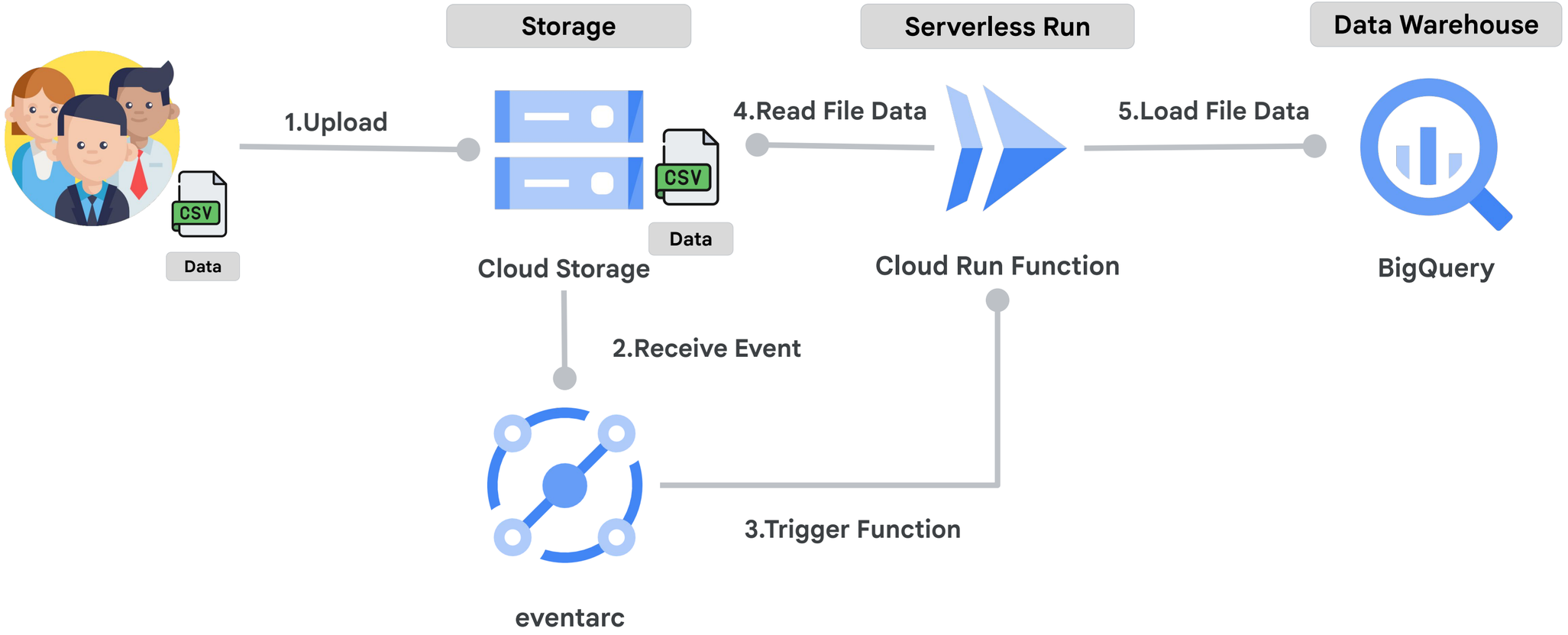


Learning objectives

- Create a Cloud Run function.
- Deploy and test the Cloud Run function.
- View data in BigQuery and review Cloud Run function logs.

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Lab (optional): Use Cloud Run Functions to Load BigQuery



Google Cloud


Google Cloud